

Medical Project(React + Django)

Functional Requirements

1. User Roles and Authentication

- **Registration & Login**
 - Patients and Doctors can register.
 - **Role-based Access Control**
 - Admin: full access to system management.
 - Doctor: manage availability and view appointments.
 - Patient: book and manage appointments.
 - **Token-based authentication** (JWT via Django Rest Framework + `django-rest-framework-simplejwt`).
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2. Admin Capabilities

- View all users (doctors & patients).
 - Approve or block doctors/patients.
 - CRUD on specialties, doctors, and other system configurations.
 - View all appointments.
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3. Doctor Capabilities

- Set availability schedule (days/times).

- View upcoming and past appointments.
 - Manage profile (specialty, bio, contact).
 - Optionally: mark appointment as approved or rejected, and can add notes.
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4. Patient Capabilities

- View list of available doctors (with filters: specialty, name, etc.).
 - View a doctor's availability.
 - Book appointment (date/time slot).
 - View, cancel, or reschedule their appointments.
 - Edit personal profile.
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5. Appointments Management

- Real-time validation of availability.
 - Avoid double booking.
 - Appointment status: pending, confirmed, completed, cancelled.
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6. Notifications

- Email confirmation for bookings.
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Technical Requirements

1. Frontend (React)

- Role-based routing using React Router.
 - Secure login/authentication handling (JWT).
 - Form validation
 - State management (Context API, Redux Toolkit).
 - UI Framework: Material UI or any other UI library except bootstrap (Don't use bootstrap)
 - API integration with Django backend.
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2. Backend (Django + Django REST Framework)

- Separate models for Users, Doctors, Patients, Appointments.
 - Use of Django's `AbstractUser` to customize roles.
 - DRF ViewSets for REST API.
 - Permissions and authentication via DRF.
 - Doctor availability stored and validated for booking.
 - Admin dashboard via Django admin.
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